

HTML HW5
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5

Given the definition of $\tilde{\mathbf{x}}_k$ and $\tilde{y}_k$, observe that

$$\mathbf{w}^T \tilde{\mathbf{x}}_k - \tilde{y}_k = w_{k-1} \sqrt{\lambda \alpha_{k-1}} - 0 = w_{k-1} \sqrt{\lambda \alpha_{k-1}}$$

for $k = 1, 2, \dots, d+1$

Hence

$$(P_V) \min_{\mathbf{w} \in \mathbb{R}^{d+1}} \frac{1}{N+K} \left(\sum_{n=1}^N (\mathbf{w}^T \mathbf{x} - y_n)^2 + \sum_{k=1}^{d+1} \lambda \alpha_{k-1} w_{k-1}^2 \right)$$

also, $N+K$ is constant with respect to different $\mathbf{w} \in \mathbb{R}^{d+1}$, so

$$\begin{aligned} \mathbf{w}^* &= \arg \min_{\mathbf{w} \in \mathbb{R}^{d+1}} \left(\sum_{n=1}^N (\mathbf{w}^T \mathbf{x} - y_n)^2 + \lambda \sum_{k=1}^{d+1} \alpha_{k-1} w_{k-1}^2 \right) \\ &= \arg \min_{\mathbf{w} \in \mathbb{R}^{d+1}} \left(\sum_{n=1}^N (\mathbf{w}^T \mathbf{x} - y_n)^2 + \lambda \sum_{i=0}^d \alpha_i w_i^2 \right) \end{aligned}$$

The solution to (P_R) is

$$\begin{aligned} &\arg \min_{\mathbf{w} \in \mathbb{R}^{d+1}} \left(\frac{1}{N} \sum_{n=1}^N (\mathbf{w}^T \mathbf{x} - y_n)^2 + \frac{\lambda}{N} \sum_{i=0}^d \alpha_i w_i^2 \right) \\ &= \arg \min_{\mathbf{w} \in \mathbb{R}^{d+1}} \left(\sum_{n=1}^N (\mathbf{w}^T \mathbf{x} - y_n)^2 + \lambda \sum_{i=0}^d \alpha_i w_i^2 \right) \quad (N \text{ is constant}) \end{aligned}$$

Therefore we see that the optimal $\mathbf{w}^*$ for (P_V) is also the optimal solution for (P_R) .
Q.E.D.

6

First, we have

$$\begin{aligned}\nabla \tilde{E}_{\text{aug}}(\mathbf{w}) &= \nabla \tilde{E}_{\text{in}} + \nabla \left(\frac{\lambda}{N} \|\mathbf{w}\|^2 \right) \\ &= \frac{1}{2} \nabla \left((\mathbf{w} - \mathbf{w}^*)^T \mathbf{H} (\mathbf{w} - \mathbf{w}^*) \right) + \frac{\lambda}{N} \nabla \|\mathbf{w}\|^2\end{aligned}$$

Then,

$$\nabla \left((\mathbf{w} - \mathbf{w}^*)^T \mathbf{H} (\mathbf{w} - \mathbf{w}^*) \right) = \left((\mathbf{w} - \mathbf{w}^*)^T \mathbf{H} \right)^T + \mathbf{H} (\mathbf{w} - \mathbf{w}^*)$$

Since the Hessian is symmetric, the expression above becomes

$$2\mathbf{H}(\mathbf{w} - \mathbf{w}^*)$$

And,

$$\nabla \|\mathbf{w}\|^2 = 2\mathbf{w}$$

Let the minimizer of $\tilde{E}_{\text{aug}}$ be $\mathbf{u}$, then

$$\begin{aligned}\nabla \tilde{E}_{\text{aug}}(\mathbf{u}) &= \mathbf{H}(\mathbf{u} - \mathbf{w}^*) + \frac{2\lambda}{N} \mathbf{u} \\ \implies \left(\mathbf{H} + \frac{2\lambda}{N} \mathbf{I}_{d+1} \right) \mathbf{u} &= \mathbf{H} \mathbf{w}^*\end{aligned}$$

($\mathbf{I}_{d+1}$ is the $d + 1$ -dimension identity matrix)

Next, we prove that $\mathbf{H} + \frac{2\lambda}{N} \mathbf{I}_{d+1}$ is invertible.

Let $\mathbf{A} = \mathbf{H} + \frac{2\lambda}{N} \mathbf{I}_{d+1}$, consider $\mathbf{v}^T \mathbf{A} \mathbf{v}$ for any non-zero vector $\mathbf{v} \in \mathbb{R}^{d+1}$

$$\begin{aligned}\mathbf{v}^T \mathbf{A} \mathbf{v} &= \mathbf{v}^T \mathbf{H} \mathbf{v} + \frac{2\lambda}{N} \mathbf{v}^T \mathbf{I}_{d+1} \mathbf{v} \\ &= \mathbf{v}^T \mathbf{H} \mathbf{v} + \frac{2\lambda}{N} \|\mathbf{v}\|^2\end{aligned}$$

Since the Hessian is positive semi-definite, $\mathbf{v}^T \mathbf{H} \mathbf{v} \geq 0$, so

$$\mathbf{v}^T \mathbf{A} \mathbf{v} \geq \frac{2\lambda}{N} \|\mathbf{v}\|^2 > 0$$

Hence $\mathbf{A}$ is positive definite, and therefore is invertible.

Given that $A = H + \frac{2\lambda}{N}I_{d+1}$ is invertible, we can express $\mathbf{u}$ as

$$\begin{aligned}
\mathbf{u} &= (H + \frac{2\lambda}{N}I_{d+1})^{-1}H\mathbf{w}^* \\
&= (H + \frac{2\lambda}{N}I_{d+1})^{-1} \left((H + \frac{2\lambda}{N}I_{d+1}) - \frac{2\lambda}{N}I_{d+1} \right) \mathbf{w}^* \\
&= \left(I_{d+1} - \frac{2\lambda}{N}(H + \frac{2\lambda}{N}I_{d+1})^{-1} \right) \mathbf{w}^* \\
&= \left(I_{d+1} - (\frac{N}{2\lambda}H + I_{d+1})^{-1} \right) \mathbf{w}^*
\end{aligned}$$

7

$$\begin{aligned}
& \mathbb{E} \left(\frac{1}{K} \sum_{n=N-K+1}^N (y_n - \bar{y})^2 \right) \\
&= \frac{1}{K} \sum_{n=N-K+1}^N \mathbb{E} (y_n^2 - 2y_n \bar{y} + \bar{y}^2) \\
&= \frac{1}{K} \sum_{n=N-K+1}^N (\mathbb{E}(y_n^2) - 2\mathbb{E}(y_n \bar{y}) + \mathbb{E}(\bar{y}^2))
\end{aligned}$$

Since $y_1, y_2, \dots, y_n$ is generated i.i.d. from a distribution with zero mean and σ^2 variance, we know

$$\mathbb{E}(y_i y_j) = \begin{cases} 0 & i \neq j \\ \sigma^2 & i = j \end{cases}$$

Now we can calculate the three terms separately

First,

$$\mathbb{E}(y_n^2) = \sigma^2$$

Second,

$$\mathbb{E}(y_n \bar{y}) = \mathbb{E} \left(y_n \frac{1}{N-K} \sum_{i=1}^{N-K} y_i \right) = \frac{1}{N-K} \mathbb{E} \left(y_n \sum_{i=1}^{N-K} y_i \right)$$

Since $n = N - K + 1, N - K + 2, \dots, N$ and $i = 1, 2, \dots, N - K$, we always have $n \neq i$, so

$$\mathbb{E}(y_n \bar{y}) = 0$$

Third,

$$\begin{aligned}
\mathbb{E}(\bar{y}^2) &= \mathbb{E} \left(\left(\frac{1}{N-K} \sum_{i=1}^{N-K} y_i \right)^2 \right) \\
&= \frac{1}{(N-K)^2} \mathbb{E} \left(\left(\sum_{i=1}^{N-K} y_i \right)^2 \right) \\
&= \frac{1}{(N-K)^2} \mathbb{E} \left(\sum_{i=1}^{N-K} y_i^2 + \sum_{i=2}^{N-K} \sum_{j=1}^{i-1} y_i y_j \right) \\
&= \frac{1}{(N-K)^2} (N-K) \sigma^2 \\
&= \frac{\sigma^2}{N-K}
\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{E}\left(\frac{1}{K}\sum_{n=N-K+1}^N(y_n - \bar{y})^2\right) &= \frac{1}{K}\sum_{n=N-K+1}^N(\sigma^2 - 2 \cdot 0 + \frac{\sigma^2}{N-K}) \\ &= \frac{1}{K}K(1 + \frac{1}{N-K})\sigma^2 \\ &= \frac{N-K+1}{N-K}\sigma^2\end{aligned}$$

$$\begin{aligned}
E_{\text{loocv}}(\mathcal{A}_{\text{avg}}) &= \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{N-1} \left(\sum_{j=1}^{i-1} y_j + \sum_{j=i+1}^N y_j \right) - y_i \right)^2 \\
&= \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{N-1} (Nw_0^* - y_i) - y_i \right)^2 \\
&= \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{N-1} (Nw_0^* - Ny_i) \right)^2 \\
&= \frac{N}{(N-1)^2} \sum_{i=1}^N (w_0^* - y_i)^2
\end{aligned}$$

Also,

$$E_{\text{in}}(w_0^*) = \frac{1}{N} \sum_{i=1}^N (w_0^* - y_i)^2$$

By comparing the expressions, we see that

$$E_{\text{loocv}}(\mathcal{A}_{\text{avg}}) = \left(\frac{N}{N-1} \right)^2 E_{\text{in}}(w_0^*)$$

Q.E.D.

9

When deployed to a test distribution $P(y = -1) = p$, E_{out} is given by

$$E_{\text{out}}(g) = p\epsilon_- + (1 - p)\epsilon_+$$

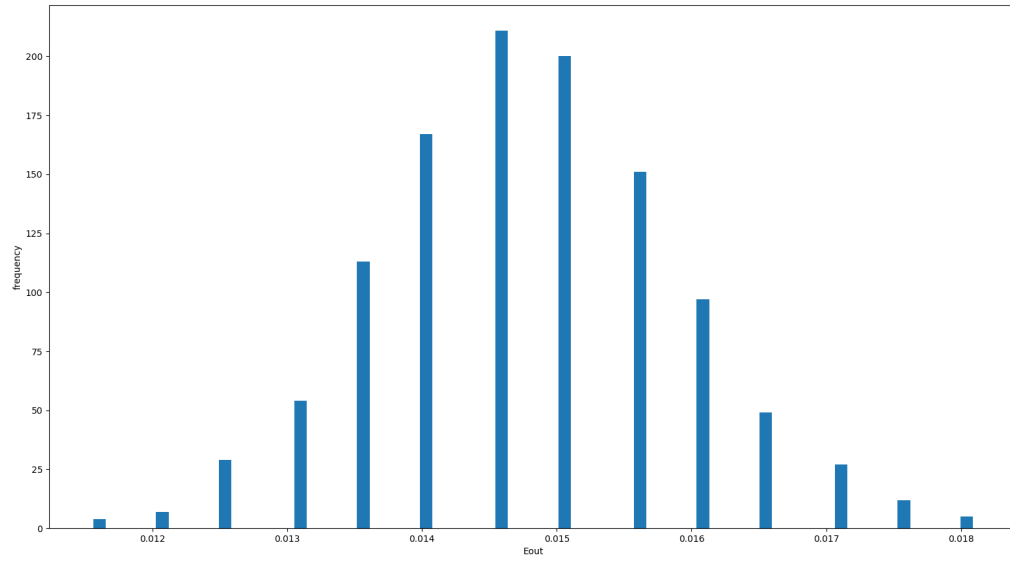
The binary classifier g is as good as the constant classifier g_c when this $E_{\text{out}}(g)$ equals $E_{\text{out}}(g_c)$, hence

$$\begin{aligned} p\epsilon_- + (1 - p)\epsilon_+ &= p \\ \implies (1 + \epsilon_+ - \epsilon_-)p &= \epsilon_+ \\ \implies p &= \frac{\epsilon_+}{1 + \epsilon_+ - \epsilon_-} \in [0, 1] \end{aligned}$$

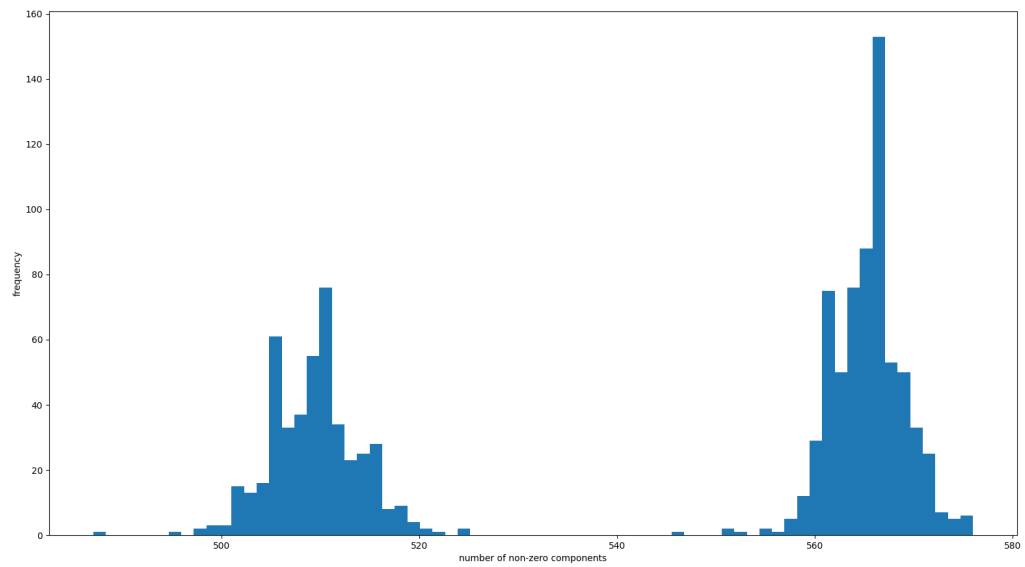
Hence when $p = \frac{\epsilon_+}{1 + \epsilon_+ - \epsilon_-}$, g is as good as g_c .

10

Histogram of $E_{\text{out}}(g)$:



Histogram of the number of non-zero components in each g :

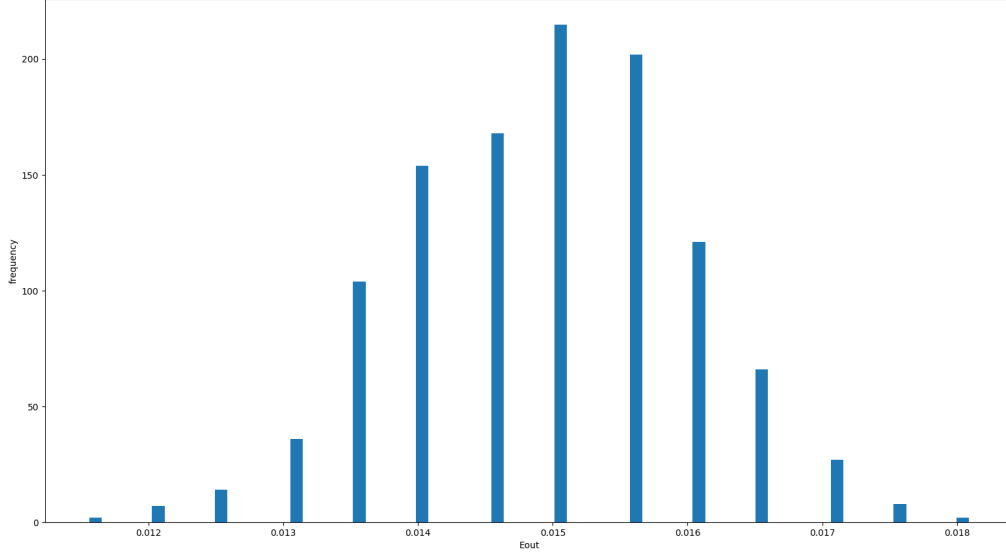


Code snapshot:

Language: `c++` (using `linhear.h`)

11

Histogram of $E_{\text{out}}(g)$:

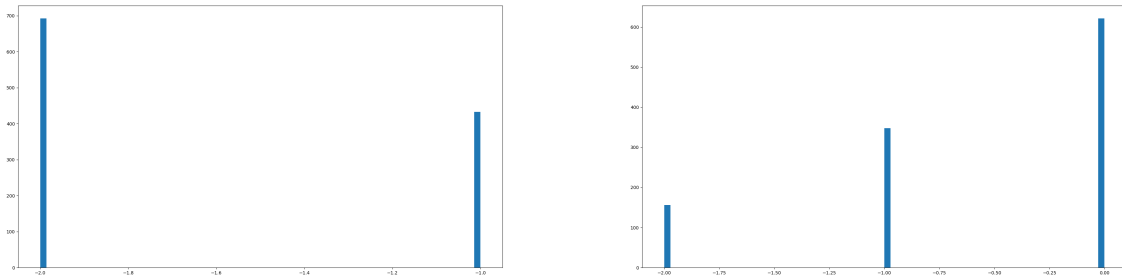


Findings:

By comparison between the distribution of E_{out} in P11 and P10, we can find that they are really similar. They both are very close to normal distribution centered at around 0.015, maxima at around 0.018 and minima at around 0.012.

Although the distribution appears to be similar, there are slight differences.

First, throughout the 1126 experiments, I recorded every $\log \lambda^*$ selected, and draw them in histograms.

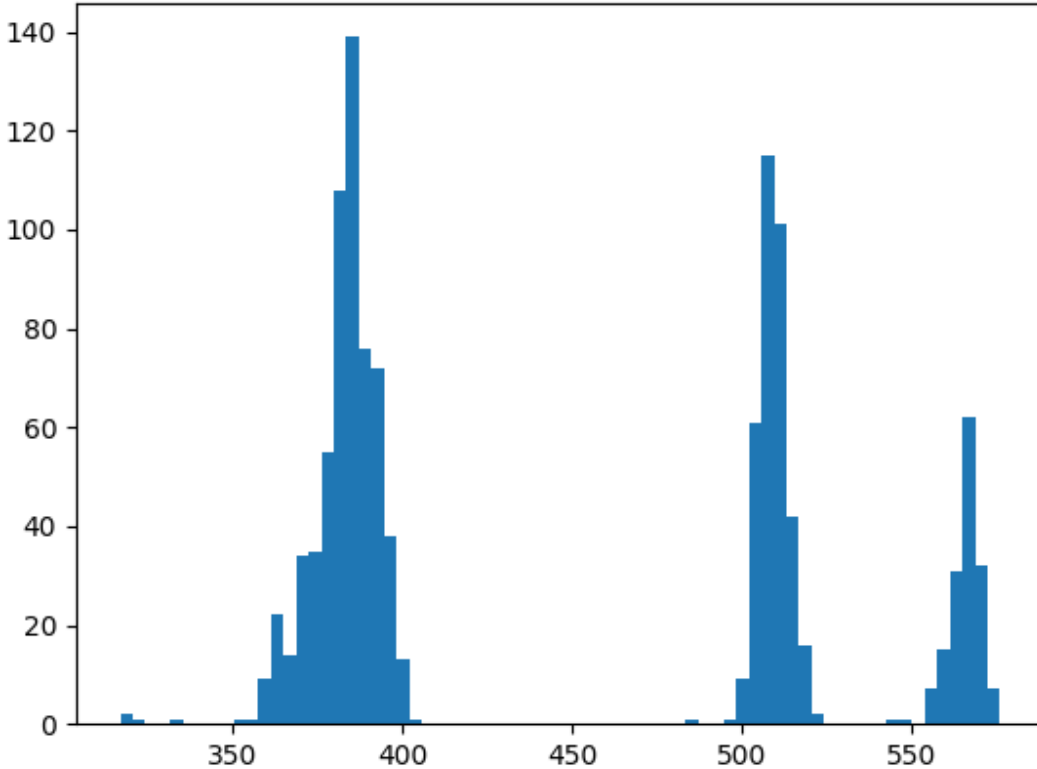


The histogram on the left is for P10, and the one on the right is for P11. From the histograms we can see that in P10 all of the λ^* are either 0.01 or 0.1, while in the major value of P11 λ^* is 1. This indicates that the validation process favors models that are more strongly regularized. A possible explanation to this phenomena is that while models that are weakly regularized has a higher dimension of freedom when fitting the training set, and hence a better E_{in} , its performance

on the validation set might not be good due to overfitting. On the other hand, by applying a higher regularization factor, overfitting can be combated and a model can perform better on the validation set.

Second, by comparing the two distributions, we see that in P11, the distribution decays faster for values greater than the center (≈ 0.015) than in P10. Also, the frequency of the largest E_{out} (around 0.018) is lower in P11 than in P10. This suggests that by choosing the model with validation, instances of high E_{out} is prevented to some extent, showing the effect of validation.

Out of curiosity, I also plot the histogram of the number of non-zero components for g in P11,

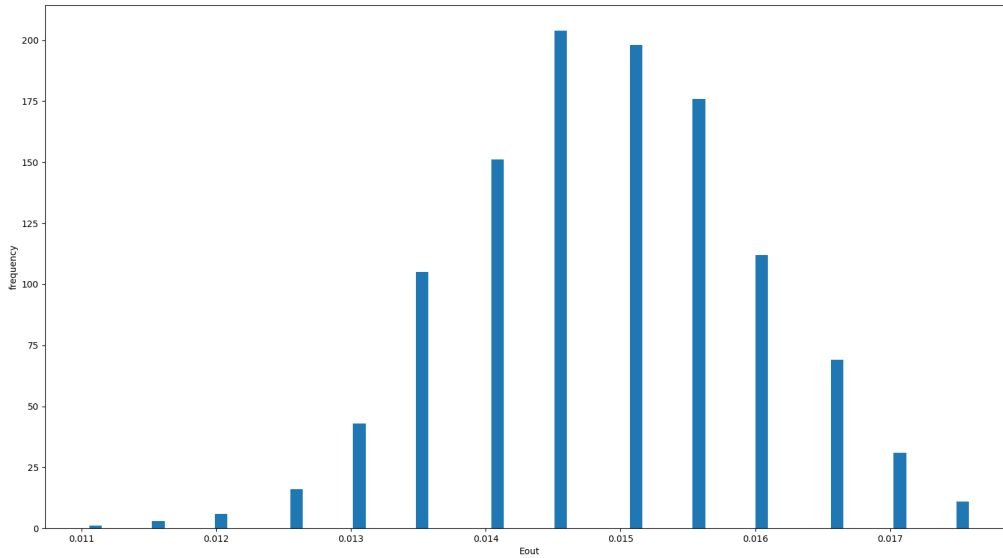


We see that compared with the plot in P10, which only has two peaks for the number of non-zero components, here there are three peaks. Together with the histograms for λ^* in P10 and P11, we can conclude that one value of λ^* corresponds to one peak in the number of non-zero components in g . This is consistent with our knowledge about regularization. As the regularization factor λ penalizes the norm of the weight vector, its value should also be related to the sparsity of the weight vector, and hence the number of non-zero components in g . Furthermore, as a higher peak should correspond to a λ value that is chosen more often, so we can conclude that a larger λ^* corresponds to a peak with a lower number of non-zero components for g . This also fits theory, as a higher

regularization factor should result in models with sparser weight vectors.
Code snapshot:

12

Histogram of $E_{\text{out}}(g)$:

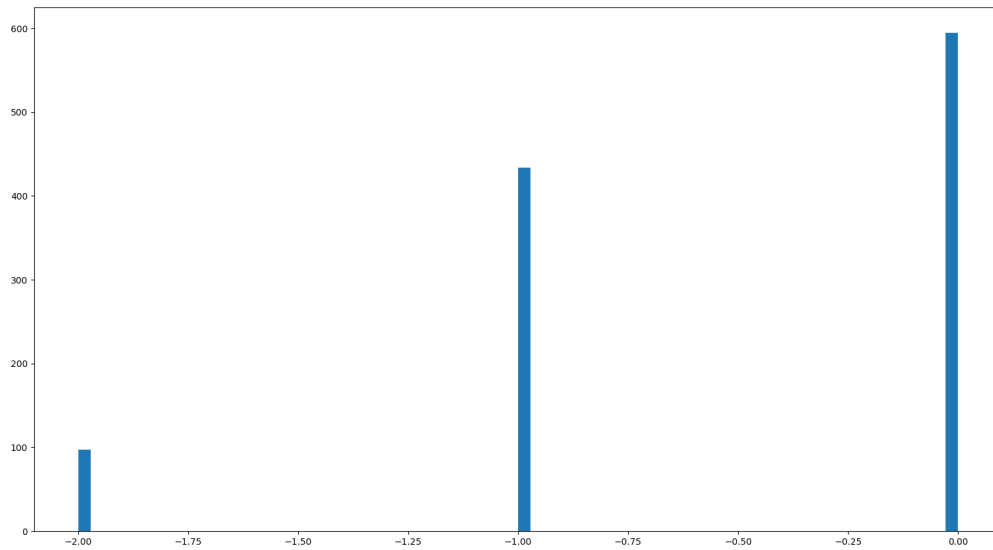


Findings :

The distribution in P12 is also quite similar to that in P11 (as well as P10), exhibiting the traits of a normal distribution centered at approximately 0.015.

However, we can still see that the value of E_{out} is smaller in P12. We see that in the distribution in P12, no samples with E_{out} around 0.018 is presented, also, the minima of E_{out} in P12 is also lower than that in P11. This suggests that for this particular data set, 3-fold cross validation might perform better than the validation method adopted in P11, in terms of selecting a good model.

I also recorded the λ^* chosen in this problem and draw them on a histogram, as shown below.



By comparing this histogram with the one for P11 (the bottom-right plot on page 10), we see that the distribution of the λ^* chosen is very similar in P11 and P12. Therefore, despite the differences in the distribution of E_{out} , we see that the choice of models by the two validation methods doesn't differ much.

Code snapshot:

13

First define the parameter $c_{n,i}$ to simplify calculation

$$c_{n,i} = y_n - \sum_{j \neq i} w_j^{(t)} x_{n,j}$$

Then we can rewrite $w_i^{(t+1)}$ as

$$w_i^{(t+1)} \leftarrow \arg \min_{w_i \in \mathbb{R}} \frac{1}{N} \sum_{n=1}^N (c_{n,i} - w_i x_{n,i})^2 + \frac{\lambda_1}{N} \left(\sum_{j \neq i} |w_j^{(t)}| + |w_i| \right) + \frac{\lambda_2}{N} \left(\sum_{j \neq i} (w_j^{(t)})^2 + w_i^2 \right)$$

Since N , $\sum_{j \neq i} |w_j^{(t)}|$ and $\sum_{j \neq i} (w_j^{(t)})^2$ are constant, $w_i^{(t+1)}$ has the simpler form

$$\begin{aligned} w_i^{(t+1)} &\leftarrow \arg \min_{w_i \in \mathbb{R}} \sum_{n=1}^N (c_{n,i} - w_i x_{n,i})^2 + \lambda_1 |w_i| + \lambda_2 w_i^2 \\ \Rightarrow w_i^{(t+1)} &\leftarrow \arg \min_{w_i \in \mathbb{R}} \left(\sum_{n=1}^N x_{n,i}^2 + \lambda_2 \right) w_i^2 + \left(\text{sign}(w_i) \lambda_1 - 2 \sum_{n=1}^N (c_{n,i} x_{n,i}) \right) w_i + \sum_{n=1}^N c_{n,i}^2 \\ \Rightarrow w_i^{(t+1)} &\leftarrow \arg \min_{w_i \in \mathbb{R}} \left(\sum_{n=1}^N x_{n,i}^2 + \lambda_2 \right) w_i^2 + \left(\text{sign}(w_i) \lambda_1 - 2 \sum_{n=1}^N (c_{n,i} x_{n,i}) \right) w_i \quad \left(\sum_{n=1}^N c_{n,i}^2 \text{ is constant} \right) \end{aligned}$$

Now we analyze the quadratic function $\left(\sum_{n=1}^N x_{n,i}^2 + \lambda_2 \right) w_i^2 + \left(\text{sign}(w_i) \lambda_1 - 2 \sum_{n=1}^N (c_{n,i} x_{n,i}) \right) w_i$. We can rewrite this in the concise form $aw_i^2 + bw_i$, where

$$\begin{aligned} a &= \left(\sum_{n=1}^N x_{n,i}^2 + \lambda_2 \right) \\ b &= \begin{cases} \lambda_1 - 2 \sum_{n=1}^N (c_{n,i} x_{n,i}) & w_i \geq 0 \\ -\lambda_1 - 2 \sum_{n=1}^N (c_{n,i} x_{n,i}) & w_i < 0 \end{cases} \end{aligned}$$

Since $\lambda_2 \geq 0$ and clearly $\sum_{n=1}^N x_{n,i}^2 \geq 0$, we have $a \geq 0$

Given that $\lambda_1 \geq 0$, the sign of b has three different cases.

Case 1 : $\frac{\lambda_1}{2} < \sum_{n=1}^N (c_{n,i} x_{n,i})$

$$\text{sign}(b) = -1$$

Case 2 : $|\sum_{n=1}^N (c_{n,i} x_{n,i})| \leq \frac{\lambda_1}{2}$

$$\text{sign}(b) = \begin{cases} +1 \vee 0 & w_i \geq 0 \\ -1 \vee 0 & w_i < 0 \end{cases}$$

(For case 2, define $\text{sign}(0) = 0$)

Case 3 : $\sum_{n=1}^N (c_{n,i} x_{n,i}) < -\frac{\lambda_1}{2}$

$$\text{sign}(b) = +1$$

In case 2, since $a \geq 0$, $bw_i \geq 0$, we see that $aw_i^2 + bw_i \geq 0$, hence the minima of $aw_i^2 + bw_i$ is 0, and $w_i = 0$ is a minimizer.

In case 1, since $b < 0$, there is a minima at $w_i > 0$, this is the minima of the quadratic function

$$\left(\sum_{n=1}^N x_{n,i}^2 + \lambda_2 \right) w_i^2 \left(\lambda_1 - 2 \sum_{n=1}^N (c_{n,i} x_{n,i}) \right) w_i$$

Hence the minimizer is

$$w_i^{(t+1)} = \frac{-\frac{\lambda_1}{2} + \sum_{n=1}^N (c_{n,i} x_{n,i})}{\sum_{n=1}^N x_{n,i}^2 + \lambda_2}$$

Similarly, in case 3, since $b > 0$, there is a minima at $w_i < 0$, this is the minima of the quadratic function

$$\left(\sum_{n=1}^N x_{n,i}^2 + \lambda_2 \right) w_i^2 \left(-\lambda_1 - 2 \sum_{n=1}^N (c_{n,i} x_{n,i}) \right) w_i$$

Hence the minimizer is

$$w_i^{(t+1)} = \frac{\frac{\lambda_1}{2} + \sum_{n=1}^N (c_{n,i} x_{n,i})}{\sum_{n=1}^N x_{n,i}^2 + \lambda_2}$$

Hence we can conclude that

$$w_i(t+1) = \begin{cases} 0 & |\sum_{n=1}^N (c_{n,i} x_{n,i})| \leq \frac{\lambda_1}{2} \\ \frac{-\frac{\lambda_1}{2} + \sum_{n=1}^N (c_{n,i} x_{n,i})}{\sum_{n=1}^N x_{n,i}^2 + \lambda_2} & \frac{\lambda_1}{2} < \sum_{n=1}^N (c_{n,i} x_{n,i}) \\ \frac{\frac{\lambda_1}{2} + \sum_{n=1}^N (c_{n,i} x_{n,i})}{\sum_{n=1}^N x_{n,i}^2 + \lambda_2} & \sum_{n=1}^N (c_{n,i} x_{n,i}) < -\frac{\lambda_1}{2} \end{cases}$$

$$= \alpha \cdot \max(0, \beta)$$

Where α and β is given by

$$\alpha = \text{sign}\left(\sum_{n=1}^N (c_{n,i} x_{n,i})\right)$$

$$\beta = \frac{-\frac{\lambda_1}{2} + |\sum_{n=1}^N (c_{n,i} x_{n,i})|}{\sum_{n=1}^N x_{n,i}^2 + \lambda_2}$$

Note that in the definition of α above, our choice for $\text{sign}(0)$ is not important, as in the case where $\sum_{n=1}^N (c_{n,i} x_{n,i}) = 0$, $\beta \leq 0$, and therefore $w_i^{(t+1)} = 0$